EXPERIMENT 9: KNOWLEDGE GRAPH CONSTRUCTION USING PYTHON

Aim:

To construct a knowledge graph of entities and relationships using a directed graph.

Software Required: Python 3.x and NetworkX.

Install NetworkX using:

pip install networkx

Procedure / Algorithm:

1. Create a directed graph.
2. Add entity relationships as labeled edges.
3. Print the subject, relation, and object for each edge.

Program:

```
knowledge_graph = [
    ("Arun", "studies", "Python"),
    ("Arun", "belongs_to", "Computer Science"),
    ("Python", "is_a", "Programming Language")
]


print("Knowledge Graph Triples:")
for subject, relation, obj in knowledge_graph:
    print(subject, "--", relation, "-->", obj)
```

Sample Input:

Entities and relationships are predefined in the program.

Expected Output:

```
==== RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-9.py ===
Knowledge Graph Triples:
Arun -- studies --> Python
Arun -- belongs_to --> Computer Science
Python -- is_a --> Programming Language
>>>
```

Result:

Thus, the knowledge graph represented entities as nodes and relationships as labeled edges.